

This document is part of the electronic companion (EC) of the paper “A Fast and Effective Method for Euclidean Anticlustering: The Assignment-Based-Anticlustering Algorithm” by Philipp Baumann, Olivier Goldschmidt, Dorit S. Hochbaum and Jason Yang. It provides additional results on the application of the ABA algorithm to Euclidean anticlustering with categories, which were not included in the main paper due to space constraints. Please refer to the main paper for details of the experimental setup (e.g., machine specifications).

Application of ABA to Euclidean Anticlustering with Categories

In some applications, it is desirable to distribute the values of a categorical feature evenly across anticlusters (see, e.g., Croella et al. [2025]). In Section 2 of the paper, we presented a variant of the ABA algorithm that ensures a balanced distribution of a categorical feature across anticlusters. In this experiment, we compare the performance of this variant with state-of-the-art benchmark algorithms from the literature. The benchmark algorithms for this comparison include the MILP model of the AVOC algorithm (see Croella et al. 2025) and the variants of the **fast-anticlustering** algorithm from Papenberg and Klau [2021] that can consider a categorical feature. Specifically, we consider the variants P-R5, P-R50, and P-R500, which use 5, 50, and 500 randomly generated exchange partners, respectively. The variant P-N5, which selects the five nearest neighbors of an object as exchange partners, cannot account for a categorical feature because it cannot guarantee that the exchange partners belong to the same category as the object itself. For the variant P-R5, P-R50, and P-R500, this requirement can be enforced with the function `generate_exchange_partners()`. To solve the MILP model of the AVOC algorithm, we used the Gurobi solver 12.0. We further include here as a baseline a variant of random partitioning that accounts for a categorical feature (Rand). We use the same five real-world datasets as in Croella et al. [2025]. For each dataset, we generated the categorical feature by applying the k -means clustering algorithm. We set the number of anticlusters to the values repored in Croella et al. [2025], i.e., four anticlusters for the dataset **Frogs**, three for the datasets **Abalone**, **Facebook**, and **Electric**, and two for the dataset **Pulsar**. The labels returned by the k -means algorithm were used as the values for the categorical feature. For each of the five datasets, Croella et al. [2025] tested five different values for the number of anticlusters, which yields 25 instances in total. We applied the MILP model and the ABA algorithm once to each instance. The three variants of the **fast-anticlustering** algorithm and the random partitioning approach were each applied three times per instance, and average metrics were reported.

Table 1 shows that ABA consistently outperforms the MILP model and Rand in solution quality. It matches the fast-clustering variants (P-R5, P-R50, P-R500) in solution quality but computes solutions up to three orders of magnitude faster. In fact, each solution is produced in a few milliseconds.

Dataset	N	D	K	Deviation from ofv ABA [%]						cpu ABA [s]	Deviation from cpu ABA [%]			
				ofv ABA	AVOC-MILP	P-R5	P-R50	P-R500	Rand		AVOC-MILP	P-R5	P-R50	P-R500
Abalone	4,177	10	4	41,759.99	-1.2553	0.0000	0.0000	0.0000	-0.0317	0.002	10,674.6	62,001.8	67,255.9	117,917.1
			5	41,759.98	-1.1358	0.0000	0.0000	0.0000	-0.0403	0.002	12,923.5	62,052.6	67,745.3	120,525.0
			6	41,759.98	-1.3272	0.0000	0.0000	0.0000	-0.0461	0.002	19,285.3	58,284.5	64,125.2	114,770.1
			8	41,759.97	-1.7711	0.0000	0.0000	0.0000	-0.0838	0.002	42,450.9	49,810.9	54,246.0	97,610.5
			10	41,759.94	-1.8276	0.0000	0.0000	0.0000	-0.1383	0.002	113,170.1	43,891.4	48,550.6	87,483.7
Facebook	7,050	13	7	91,636.97	-5.3732	0.0000	0.0000	0.0000	-0.0564	0.003	21,057.7	162,745.9	170,773.2	210,257.3
			8	91,636.83	-5.7497	0.0000	0.0000	0.0000	-0.0721	0.003	31,800.2	155,585.5	166,008.6	201,412.7
			10	91,636.56	-5.9977	0.0000	0.0000	0.0000	-0.1047	0.003	62,758.4	135,299.4	142,275.9	176,330.6
			13	91,636.56	-7.2956	-0.0001	0.0000	0.0000	-0.1355	0.004	230,917.9	109,088.5	115,277.2	142,208.9
			18	91,635.42	-7.6534	-0.0004	0.0000	0.0000	-0.1819	0.005	694,918.4	86,441.2	91,161.3	113,085.2
Frogs	7,195	22	8	158,267.92	-7.0749	0.0000	0.0000	0.0000	-0.0230	0.003	92,477.9	83,810.8	92,164.3	146,892.0
			10	158,267.86	-10.6908	0.0000	0.0001	0.0001	-0.0290	0.004	190,892.7	64,936.5	71,885.6	115,655.0
			13	158,267.77	-11.0695	-0.0001	0.0001	0.0001	-0.0380	0.005	565,602.4	57,763.8	63,695.7	104,303.4
			15	158,267.73	-12.4326	-0.0002	0.0001	0.0001	-0.0512	0.005	1,032,629.8	50,736.5	56,089.3	92,309.0
			16	158,267.67	-11.1633	-0.0002	0.0001	0.0001	-0.0490	0.005	1,488,430.2	48,157.9	53,626.8	87,766.3
Electric	10,000	12	10	119,987.95	-0.0837	0.0000	0.0000	0.0000	-0.0807	0.004	71,594.5	95,408.6	105,213.4	139,266.3
			15	119,987.90	-0.1197	0.0000	0.0000	0.0000	-0.1346	0.006	331,436.3	76,148.2	84,223.1	111,355.9
			20	119,987.83	-0.1761	-0.0001	0.0000	0.0001	-0.1889	0.007	603,941.7	58,501.0	64,829.7	86,725.9
			25	119,987.78	-0.2124	-0.0001	0.0000	0.0001	-0.2505	0.009	2,731,703.5	48,312.9	53,153.6	71,429.4
			30	119,987.71	-0.2500	-0.0003	0.0000	0.0001	-0.2947	0.010	8,400,840.5	41,976.9	46,230.5	61,967.7
Pulsar	17,898	8	18	143,176.00	-0.7841	0.0000	0.0000	0.0000	-0.0775	0.015	46,765.3	155,314.3	158,364.8	171,143.6
			20	143,176.00	-0.7946	0.0000	0.0000	0.0000	-0.0951	0.016	61,015.3	143,480.4	147,193.5	159,381.6
			25	143,176.00	-1.0012	0.0000	0.0000	0.0000	-0.1160	0.021	87,135.8	111,041.4	113,488.9	122,974.5
			30	143,175.99	-1.0790	0.0000	0.0000	0.0000	-0.1452	0.025	154,599.0	93,264.9	95,536.1	103,859.7
			35	143,175.99	-1.2821	-0.0001	0.0000	0.0000	-0.1600	0.030	232,183.3	78,080.6	79,395.5	86,167.5
			Average					-3.9040	-0.0001	0.0000	0.0000	-0.1050		693,248.2

ABA is the contribution in this paper

Table 1: Comparing solution quality and running times of ABA, P-R5, P-R50, P-R500, and random for instances with a categorical feature. Reported are the objective function values (ofv) for the ABA algorithm and the percentage deviations from these values for the other algorithms (positive deviation means improving on ABA). Also reported are the running times of the ABA algorithm in seconds (cpu) and the percentage deviations from these values for the other algorithms (negative deviation means improving on ABA).

We also calculated for each solution the diversity for each of the K anticlusters separately and then based

on these K values calculated the standard deviation (sd) and the range (difference between maximum and the minimum). Table 2 reports both metrics for all 25 instances. Consistent with the other results reported in the main paper, the ABA algorithm produces anticlusters with more balanced diversity than the other methods.

Deviation from sd ABA [%]										Deviation from range ABA [%]					
Dataset	<i>N</i>	<i>D</i>	<i>K</i>	sd ABA	AVOC-MILP	P-R5	P-R50	P-R500	Rand	range ABA	AVOC-MILP	P-R5	P-R50	P-R500	Rand
Abalone	4,177	10	4	204.4	116.9	28.6	11.8	70.6	33.6	501.8	126.9	38.3	14.6	87.7	28.0
			5	190.6	94.5	23.5	33.8	42.4	54.2	498.4	85.5	32.6	38.2	52.6	64.9
			6	177.8	118.5	54.5	52.2	59.2	31.1	496.6	132.4	76.9	66.4	75.6	31.6
			8	165.9	64.9	38.2	46.1	51.0	32.5	526.6	73.3	45.8	59.2	67.9	38.9
			10	148.9	90.1	14.7	35.3	53.9	35.5	512.2	98.1	25.4	40.7	73.5	48.2
Facebook	7,050	13	7	470.9	749.5	151.1	124.6	100.9	174.4	1,320.1	710.2	147.7	122.1	122.8	215.0
			8	463.7	660.5	216.9	129.4	167.3	161.8	1,299.2	757.7	289.3	158.6	194.9	228.7
			10	376.2	740.7	230.5	238.3	188.4	254.2	1,069.7	897.3	302.8	298.0	236.1	309.2
			13	450.6	497.8	88.8	103.3	116.3	160.1	1,527.9	493.4	102.3	101.4	126.1	197.1
			18	427.0	455.4	94.1	99.6	82.9	118.0	1,458.0	442.6	112.8	103.9	97.6	147.4
Frogs	7,195	22	8	44.1	9,162.3	591.6	799.6	760.4	1,008.0	152.3	7,162.4	524.9	730.2	665.6	942.6
			10	58.4	5,152.0	352.4	405.2	370.5	552.0	213.0	4,679.1	289.1	352.5	325.6	459.0
			13	52.6	4,206.2	375.0	422.8	379.7	515.4	211.4	3,163.6	334.3	371.4	334.5	392.1
			15	53.1	3,863.6	330.3	291.3	297.3	502.7	186.8	3,601.9	317.8	292.3	345.8	470.9
			16	54.0	3,666.5	308.9	306.1	319.1	510.5	212.0	2,933.6	275.4	319.5	270.5	416.5
Electric	10,000	12	10	2.1	4,127.7	5,608.5	5,966.3	6,039.4	4,550.0	8.0	3,450.2	4,476.3	4,795.5	5,255.0	4,146.5
			15	2.6	2,252.6	3,231.1	3,617.7	3,197.0	2,510.2	9.4	2,009.0	2,991.9	3,018.1	2,789.2	2,792.4
			20	3.0	2,121.5	2,611.9	2,737.8	2,385.0	2,457.0	12.7	1,653.9	2,095.1	2,066.9	1,981.3	2,188.8
			25	2.4	2,380.2	2,689.3	2,543.3	2,916.0	2,261.7	7.6	3,217.2	3,535.9	3,285.1	3,826.4	2,807.4
			30	2.6	2,071.2	2,142.2	2,230.8	2,229.5	2,383.7	11.4	1,680.0	2,218.1	2,087.9	2,156.5	2,141.9
Pulsar	17,898	8	18	11.1	6,271.4	1,011.7	953.0	1,131.5	3,817.7	43.1	5,574.6	919.4	1,042.3	1,226.1	3,595.0
			20	19.6	2,920.8	470.3	485.5	531.0	1,820.1	81.0	3,114.9	338.7	345.5	481.6	1,686.8
			25	21.5	2,473.3	300.4	265.8	373.2	1,990.1	85.0	2,748.9	367.1	302.3	447.7	1,859.1
			30	24.3	1,740.2	293.5	281.6	292.8	1,587.0	114.1	1,842.1	269.9	226.1	239.4	1,448.1
			35	26.7	1,450.3	248.8	210.6	217.1	1,227.4	101.7	1,675.4	278.7	238.0	247.2	1,274.4
Average					2,297.9	860.3	895.7	894.9	1,150.0		2,093.0	816.3	819.1	869.1	1,117.2

ABA is the contribution in this paper

Table 2: Comparing the statistics of the solutions produced by ABA, P-R5, P-R50, P-R500 and random. Reported are the standard deviations (sd) of the diversity values for the ABA algorithm and the percentage deviations from these values for the other algorithms (negative deviation means improving on ABA). Also reported are the ranges (range) of the diversity values for the ABA algorithm and the percentage deviations from these values for the other algorithms (negative deviation means improving on ABA).

References

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